MILITARY SPECIFICATION SHEET

PLASTIC SHEETS, LAMINATED, THERMOSETTING ELECTRICAL-

INSULATING SHEET, POLYESTER GLASS-MAT GRADE GPO-N3

The complete requirements for procuring the plastic sheet described herein shall consist of this document and the latest issue of Specification MIL-P-24364(SHIPS).

The plastic laminated sheets shall be as follows:

Grade GPO-N3 Polyester Resin Glass-Mat, Track Resistance, Low Flammability and Self-extinguishing Properties. (Temperature rating of Grade GPO-N3 plastic, shall be predicted on service experience or long term data furnished by the manufacturer and must be acceptable to the Naval Ship Engineering Center, Materials Development and Application Office.)

Requirement: The property value requirements shall be as shown in tables I and II.

FSC 9330

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Table I - Property value requirements for

Qualification test for polyester glass-mat grade GPO-N3

Properties to be tested	Test paragraph	Numoer of specimens for each thickness	Conditioning (see 4.6)	Unit of value	Values required for each thickness		
					1/16 inch	1/8 inch	1/2 inch
Thickness	4.7.3	1	As received	Inch	±0.008	±0.010	±0.024
Dielectric breakdown parallel to laminations	4.7.5	1 (short-time) 4 (step-by-step)	As received	Average KV-min.	40	40	40
		1 (short-time) 4 (step-by-step)	336/50 water	Average KV-min.	15	15	15
Dielectric constant	4.7.6			Maximum			
60 hertz		4	As received		----	6.0	----
		4	24/23 water		----	7.0	----
1 megahertz		4	As received		----	5.0	----
		4	24/23 water		----	6.0	----
Dissipation factor	4.7.6			Maximum			
60 hertz		4	As received		----	0.05	----
		4	24/23 water		----	0.15	----
1 megahertz		4	As received		----	0.03	----
		4	24/23 water		----	0.06	----
Arc resistance	4.7.7	4	As received	Seconds-min.	180	180	180
			48/50 water		180	180	180
Wet tracking	4.7.8	4	As received	Minutes-min.	300 1/	300 1/	300
Impact strength, tested edgewise	4.7.9	4	48/50	Foot-lbs. per inch-min.	----	8	8
Flexural strength, tested flatwise, (crosswise)	4.7.10	5	As received	PSI-min.	16000	18000	18000
Bond strength	4.7.11	4	As received	lbs.-min.	----	----	850
		4	48/50 water		----	----	800
Water absorption	4.7.12	4	1/105 + 24/23 water	Percent-max	0.6	0.5	0.25
Flame resistance	4.7.13	5	As received				
Ignition time				Seconds-min.	----	----	75
Burning time				Seconds-max.	----	----	85
Silicone content	4.7.15	1	168/185	p.p.m. max.	25	25	25
Machinability	4.7.16	1	As received	Pass or fail	----	----	Pass

1/ Specimens of the same material shall be built up to 0.250 inch.

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Table II - Property value requirements for

Quality conformance test for polyester glass-mat grade GPC-N3.

Properties to be tested	Test paragraph	Number of specimens for each thickness	Conditioning (see 4.6)	Units of value	Values required for each thickness range[1] (inches)				
					0.0625 or less	0.0626-0.094	0.095-0.250	0.251-1.00	1.001-2.000
Warp or twist	4.7.4	1	As received	Percent max.	2.50%	----	2.50%	0.50%	----
Dielectric breakdown parallel to laminations	4.7.5	1 (short-time) 4 (step-by-step)	48/50 water	Average KV-min.	15	15	15	15	15
Wet tracking	4.7.8[2]	4	As received	Minutes-min.	300	300	300	300	300
Flexural strength, tested, (flat-wise, (cross-wise)	4.7.10	4	As received	PSI-min.	16000	16000	18000	16000	14000
Bond strength	4.7.11	4	As received	Pounds-min.	----	----	----	850[3]	850
Flame resistance ignition time	4.7.13	5	As received	Seconds-min.	----	----	----	75	----
burning time				Seconds-max.	----	----	----	85	----

[1] These ranges applicable for the nominal thickness.

[2] Specimens of the same material shall be built up to 0.250 inches.

[3] For specimens 0.500 inches and above.

Preparing activity:
Navy - SH
(Project 9330-N447)

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